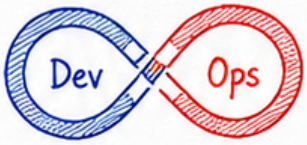
100 DevOps Engineer Interview Questions and Answers

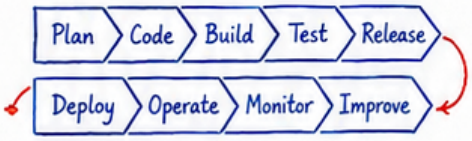
DevOps Fundamentals

1. **What is DevOps?**
   → DevOps is a culture and set of practices that combines development and operations to deliver software faster and more reliably.

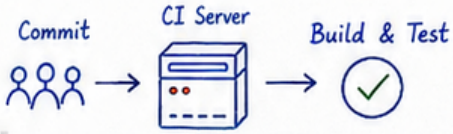


2. **What are the main goals of DevOps?**
   → Automation, faster delivery, improved collaboration, reliability, scalability, and continuous feedback.

3. **What is the DevOps lifecycle?**
   → Plan, code, build, test, release, deploy, operate, monitor, and improve.



4. **What is Continuous Integration?**
   → Continuous Integration automatically builds and tests code whenever developers commit changes.



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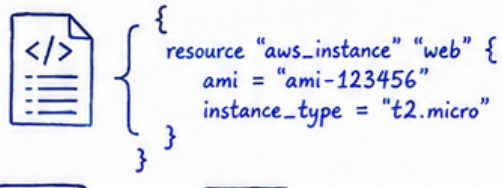
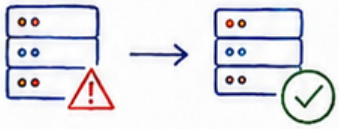
5. **What is Continuous Delivery?**
   → Continuous Delivery keeps software ready for production deployment through automated build, test, and release processes.

6. **What is Continuous Deployment?**
   → Continuous Deployment automatically releases every validated change to production.

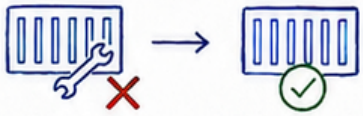
7. **What is Infrastructure as Code?**
   → Infrastructure as Code manages infrastructure through version-controlled configuration files.



8. **What is configuration management?**
   → Configuration management keeps servers and systems in a consistent, desired state.



9. **What is immutable infrastructure?**
   → Immutable infrastructure replaces servers or containers instead of modifying them after deployment.



10. **What is shift-left testing?**
    → Shift-left testing performs testing earlier in the software development lifecycle.

100 DevOps Engineer Interview Questions and Answers

Linux and Networking

11. How do you check disk usage in Linux?
→ Use df -h for filesystem usage and du -sh for directory usage.

```
$ df -h
Filesystem  Size  Used  Avail  Use%  Mounted on
/dev/sda1   50G   20G   28G    42%   /
...
```

12. How do you check memory usage in Linux?
→ Use free -m, top, or htop.

13. How do you find running processes?
→ Use ps aux, top, or htop.

```
$ ps aux | head
user  1234  0.1  1.2  12345  ?  Ss  10:11  0:00  nginx
user  2345  0.0  0.5  6789   ?  S   10:11  0:00  sshd
...
```

14. How do you stop a Linux process?
→ Use kill <PID> or kill -9 <PID> when graceful termination fails.

```
$ kill 1234
```

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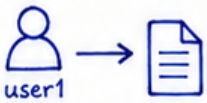
15. What does the chmod command do?
→ It changes file and directory permissions.

chmod 755 file.sh
-rwxr-xr-x

16. What does the chown command do?
→ It changes the owner and group of a file or directory.

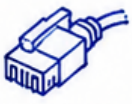


chown user1:dev file.txt

17. What is a symbolic link?
→ A symbolic link is a file that points to another file or directory.

18. How do you check listening ports?
→ Use ss -tulpn or netstat -tulpn.

```
$ ss -tulpn
Netid  State    Recv-Q  Send-Q  Local Address:Port  Process
tcp    LISTEN   0       128     0.0.0.0:22          1234/sshd
udp    UNCONN   0       0       0.0.0.0:53          987/dnsmasq
...
```



19. What is DNS?
→ DNS converts domain names into IP addresses.

20. What is the difference between TCP and UDP?
→ TCP is reliable and connection-oriented, while UDP is faster and connectionless.

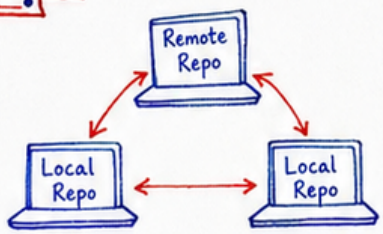
TCP	UDP
• Reliable	• Faster
• Connection-oriented	• Connectionless
• Slower	• No guarantee
• Uses handshake (SYN, ACK)	• No handshake

100 DevOps Engineer Interview Questions and Answers

Git and Version Control

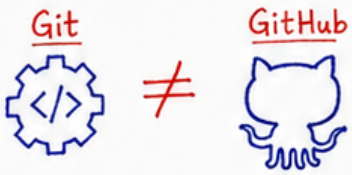
21. What is Git?

→ Git is a distributed version-control system used to track source-code changes.



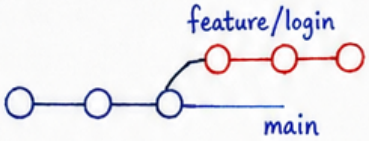
22. What is the difference between Git and GitHub?

→ Git is a version-control tool, while GitHub is a platform that hosts Git repositories.



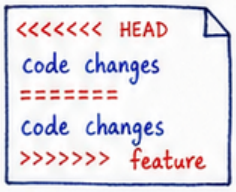
23. What is a Git branch?

→ A branch is an independent line of development.



24. What is a merge conflict?

→ A merge conflict occurs when Git cannot automatically combine conflicting changes.



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25. What is git rebase?

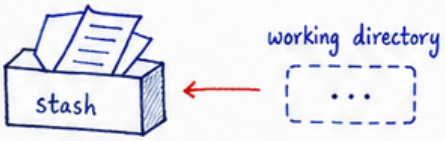
→ Rebase moves or reapplies commits onto a new base commit.

26. What is the difference between merge and rebase?

→ Merge preserves branch history, while rebase creates a cleaner linear history.

27. What does git stash do?

→ It temporarily saves uncommitted changes without creating a commit.

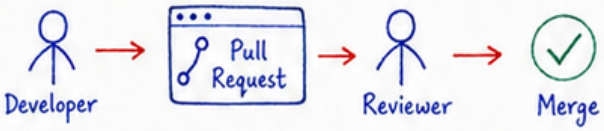


28. What does git cherry-pick do?

→ It applies a specific commit from one branch to another.

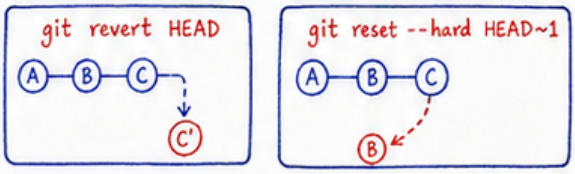
29. What is a pull request?

→ A pull request proposes code changes for review before merging.



30. How do you undo the latest Git commit?

→ Use git revert for a safe reversal or git reset to move the branch pointer.



100 DevOps Engineer Interview Questions and Answers

CI/CD

31. What is a CI/CD pipeline?

→ It is an automated workflow that builds, tests, and deploys software.

32. What are common CI/CD tools?

→ Jenkins, GitHub Actions, GitLab CI/CD, Azure DevOps, CircleCI, and Argo CD.

33. What is Jenkins?

→ Jenkins is an automation server used to build, test, and deploy applications.

34. What is a Jenkins pipeline?

→ A Jenkins pipeline defines CI/CD stages as code, usually in a Jenkinsfile.

35. What is a build artifact?

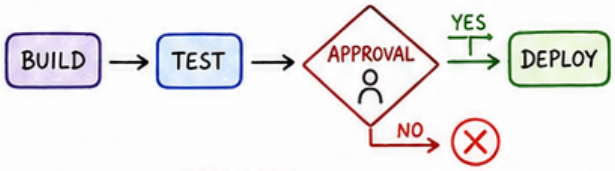
→ A build artifact is an output package such as a JAR, binary, image, or archive.

36. What is pipeline as code?

→ Pipeline as code stores CI/CD workflow definitions in version control.

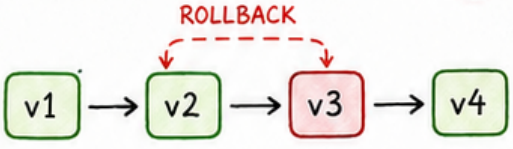
37. What is a deployment gate?

→ A deployment gate is a condition or approval required before a release proceeds.



38. What is a rollback?

→ A rollback restores an application to a previously stable version.

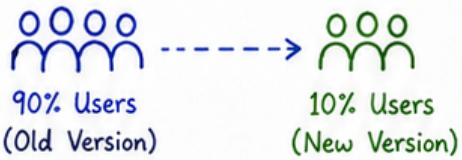


39. What is a blue-green deployment?

→ It uses two identical environments and switches traffic from the old version to the new one.

40. What is a canary deployment?

→ It releases a new version to a small percentage of users before wider deployment.



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100 DevOps Engineer Interview Questions and Answers

Docker and Containers

41. What is Docker?
→ Docker is a platform for building, packaging, and running applications in containers.

42. What is a container?
→ A container is an isolated process that packages an application with its dependencies.

43. What is a Docker image?
→ A Docker image is a read-only template used to create containers.

44. What is a Dockerfile?
→ A Dockerfile contains instructions for building a Docker image.

45. What is Docker Compose?
→ Docker Compose defines and runs multi-container applications using a YAML file.

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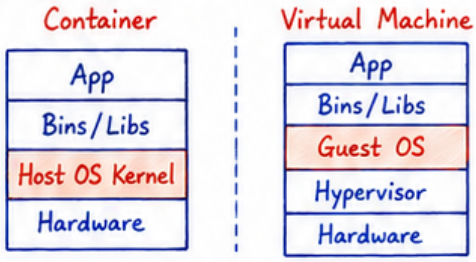
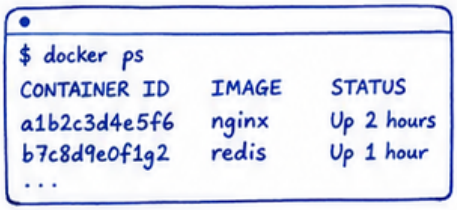


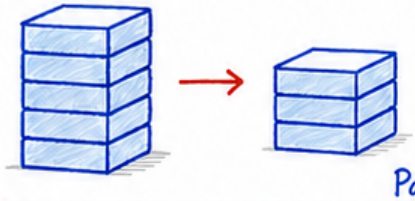
46. What is the difference between a container and a virtual machine?
→ Containers share the host kernel, while virtual machines include a complete guest operating system.

47. What is a Docker volume?
→ A Docker volume provides persistent storage for container data.

48. What is port mapping in Docker?
→ Port mapping connects a host port to a container port.



49. How do you list running Docker containers?
→ Use docker ps.



50. How do you reduce Docker image size?
→ Use small base images, multi-stage builds, .dockerignore, and fewer layers.

100 DevOps Engineer Interview Questions and Answers

Kubernetes

51. What is Kubernetes?
→ Kubernetes is a platform for deploying, scaling, and managing containerized applications.

52. What is a Kubernetes Pod?
→ A Pod is the smallest deployable Kubernetes unit and contains one or more containers.

53. What is a Kubernetes Deployment?
→ A Deployment manages application replicas and rolling updates.

54. What is a Kubernetes Service?
→ A Service provides stable network access to a group of Pods.

55. What is a Kubernetes Namespace?
→ A Namespace logically separates resources within a cluster.

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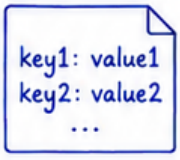
56. What is a ConfigMap?
→ A ConfigMap stores non-sensitive application configuration.



57. What is a Kubernetes Secret?
→ A Secret stores sensitive values such as passwords, tokens, and certificates.

58. What is an Ingress?
→ Ingress manages external HTTP and HTTPS access to cluster services.

59. What is a StatefulSet?
→ A StatefulSet manages stateful applications that require stable identities and storage.

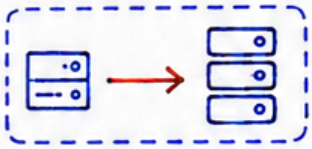
60. What is a DaemonSet?
→ A DaemonSet ensures that a Pod runs on every selected Kubernetes node.

Cloud Computing

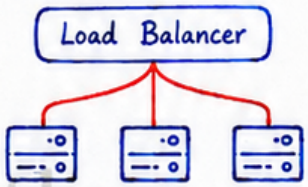
61. What is cloud computing?
→ Cloud computing provides on-demand computing resources over the internet.

62. What are IaaS, PaaS, and SaaS?
→ They are infrastructure, platform, and software services delivered through the cloud.

63. What is auto-scaling?
→ Auto-scaling automatically adjusts resources according to workload demand.



64. What is a load balancer?
→ A load balancer distributes incoming traffic across multiple servers.

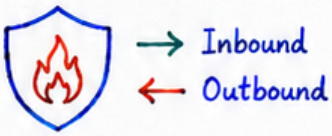


65. What is a cloud region?
→ A region is a geographic area containing cloud data centers.

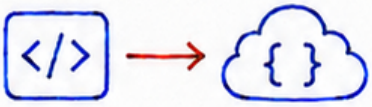
66. What is an availability zone?
→ An availability zone is an isolated data-center location within a region.

67. What is a virtual private cloud?
→ A VPC is a logically isolated network inside a public cloud.

68. What is a security group?
→ A security group is a virtual firewall controlling inbound and outbound traffic.



69. What is serverless computing?
→ Serverless computing runs code without requiring users to manage servers.



70. What is the shared-responsibility model?
→ The cloud provider secures the cloud, while customers secure their workloads and data.

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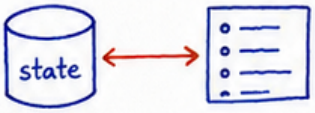
Infrastructure as Code and Automation

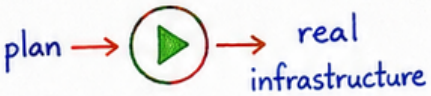
71. **What is Terraform?**
→ Terraform is an Infrastructure as Code tool used to provision and manage infrastructure.

72. **What is Terraform state?**
→ Terraform state maps configuration resources to real infrastructure.

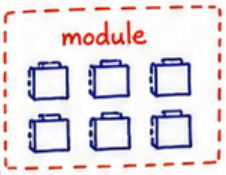


73. **What does terraform plan do?**
→ It previews the infrastructure changes Terraform intends to make.

74. **What does terraform apply do?**
→ It creates or updates infrastructure according to the Terraform configuration.



75. **What is a Terraform module?**
→ A module is a reusable collection of Terraform resources.



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76. **What is Ansible?**
→ Ansible is an agentless automation and configuration-management tool.

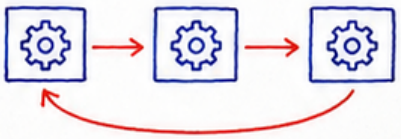
77. **What is an Ansible playbook?**
→ An Ansible playbook is a YAML file that defines automation tasks.

```
playbook.yml
- hosts: web
  tasks:
  - name: install nginx
    apt: name=nginx
      state=present
```

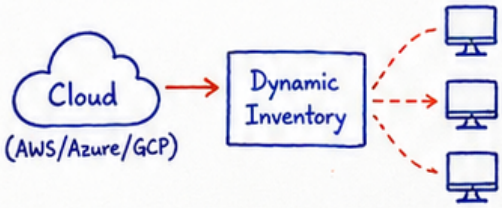
78. **What is an Ansible inventory?**
→ An inventory defines the hosts and groups managed by Ansible.

```
[web]
web1.example.com
web2.example.com
[db]
db1.example.com
```

79. **What is idempotency?**
→ Idempotency means repeated execution produces the same desired result without unnecessary changes.



80. **What is a dynamic inventory?**
→ A dynamic inventory automatically retrieves hosts from cloud platforms or external systems.



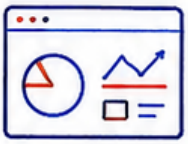
100 DevOps Engineer Interview Questions and Answers

Monitoring and Logging

81. What is monitoring? → Monitoring collects and analyzes system metrics to identify health and performance issues.

82. What is observability? → Observability uses metrics, logs, and traces to understand a system's internal state.

83. What is Prometheus? → Prometheus is a monitoring system that collects and queries time-series metrics.



84. What is Grafana? → Grafana creates dashboards and visualizations from monitoring data.

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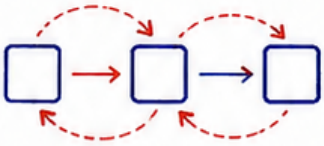
85. What is the ELK Stack? → ELK combines Elasticsearch, Logstash, and Kibana for centralized log management.

86. What is an alert? → An alert notifies teams when a metric or condition crosses a defined threshold.

87. What is application performance monitoring? → APM tracks application response times, errors, dependencies, and transaction performance.



88. What is distributed tracing? → Distributed tracing tracks a request as it moves through multiple services.

89. What is an SLA? → An SLA is a formal agreement defining expected service availability or performance.

90. What are SLI and SLO? → An SLI measures service performance, while an SLO defines the target value.

100 DevOps Engineer Interview Questions and Answers

Security and Troubleshooting

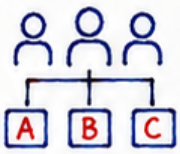
91. What is DevSecOps?
→ DevSecOps integrates security practices throughout the DevOps lifecycle.

92. What is the principle of least privilege?
→ Users and services receive only the permissions necessary to perform their tasks.

93. How should secrets be managed?
→ Store them in a secure secrets manager and never hard-code them in source code.

94. What is vulnerability scanning?
→ Vulnerability scanning checks code, dependencies, images, and systems for known security weaknesses.

95. What is RBAC?
→ Role-Based Access Control grants permissions according to assigned roles.



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96. How do you troubleshoot a failed deployment?
→ Check pipeline logs, application logs, events, configuration, resource limits, and recent changes.

97. How do you troubleshoot high CPU usage?
→ Identify the consuming process, inspect logs and metrics, and check traffic or application behavior.

98. How do you troubleshoot a Kubernetes CrashLoopBackOff error?
→ Inspect Pod logs, events, commands, environment variables, probes, and resource limits.

99. What is root-cause analysis?
→ Root-cause analysis identifies the underlying reason an incident occurred.

100. What is a post-incident review?
→ It is a blameless review that documents the incident, impact, root cause, lessons, and preventive actions.